Voiding Dysfunction

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Conflicts of Interest

- None

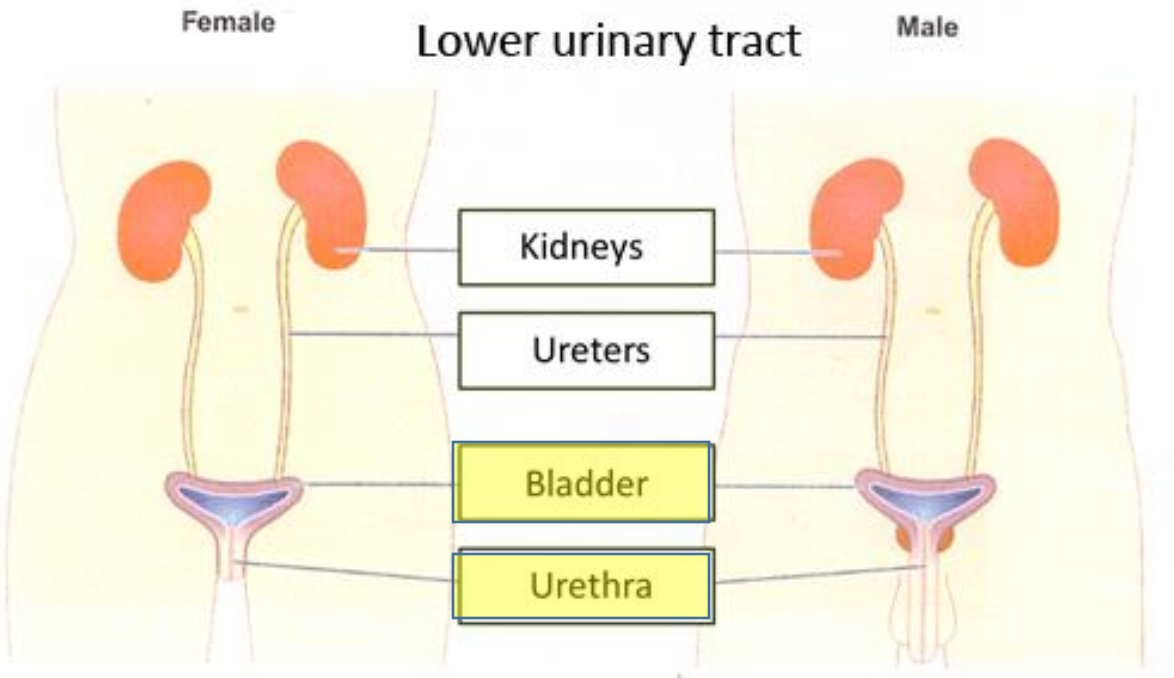
Learning Objectives

1. Identify the components of the lower urinary tract and describe their functional role
2. Describe the pathways for lower urinary tract sympathetic, parasympathetic and somatic innervation
3. Explain the difference between stress and urgency incontinence
4. Describe the nature of lower urinary tract dysfunction based on the location of a neurologic lesion
5. Describe the etiologies for bladder outlet obstruction and identify how they are treated

Overview

- Lower urinary tract anatomy
- Normal lower urinary tract physiology
- Urinary incontinence
  - Stress incontinence
  - Urgency incontinence
- Neurogenic bladder
- Bladder outlet obstruction

Lower Urinary Tract Anatomy

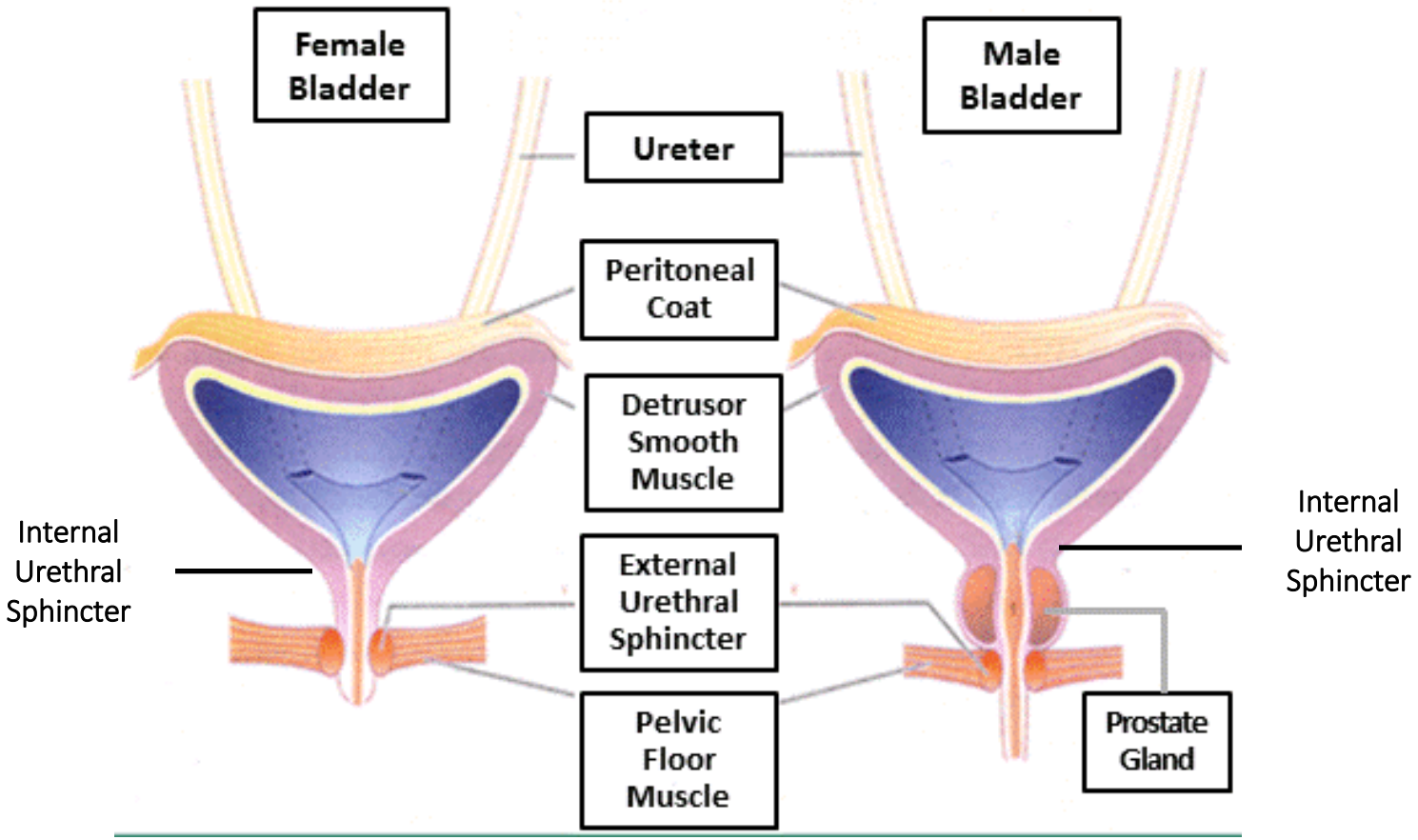


Lower Urinary Tract Anatomy

1. Bladder body
   - Dynamic allowance of filling/emptying
   - Smooth muscle bundles oriented in random directions

2. Bladder outlet
   - Trigone
     - Ureteral insertion/drainage
     - Smooth muscle bundles in longitudinal + circular orientation
   - Urethra
     - Bladder neck = internal urethral sphincter
     - External urethral sphincter

Lower Urinary Tract Anatomy



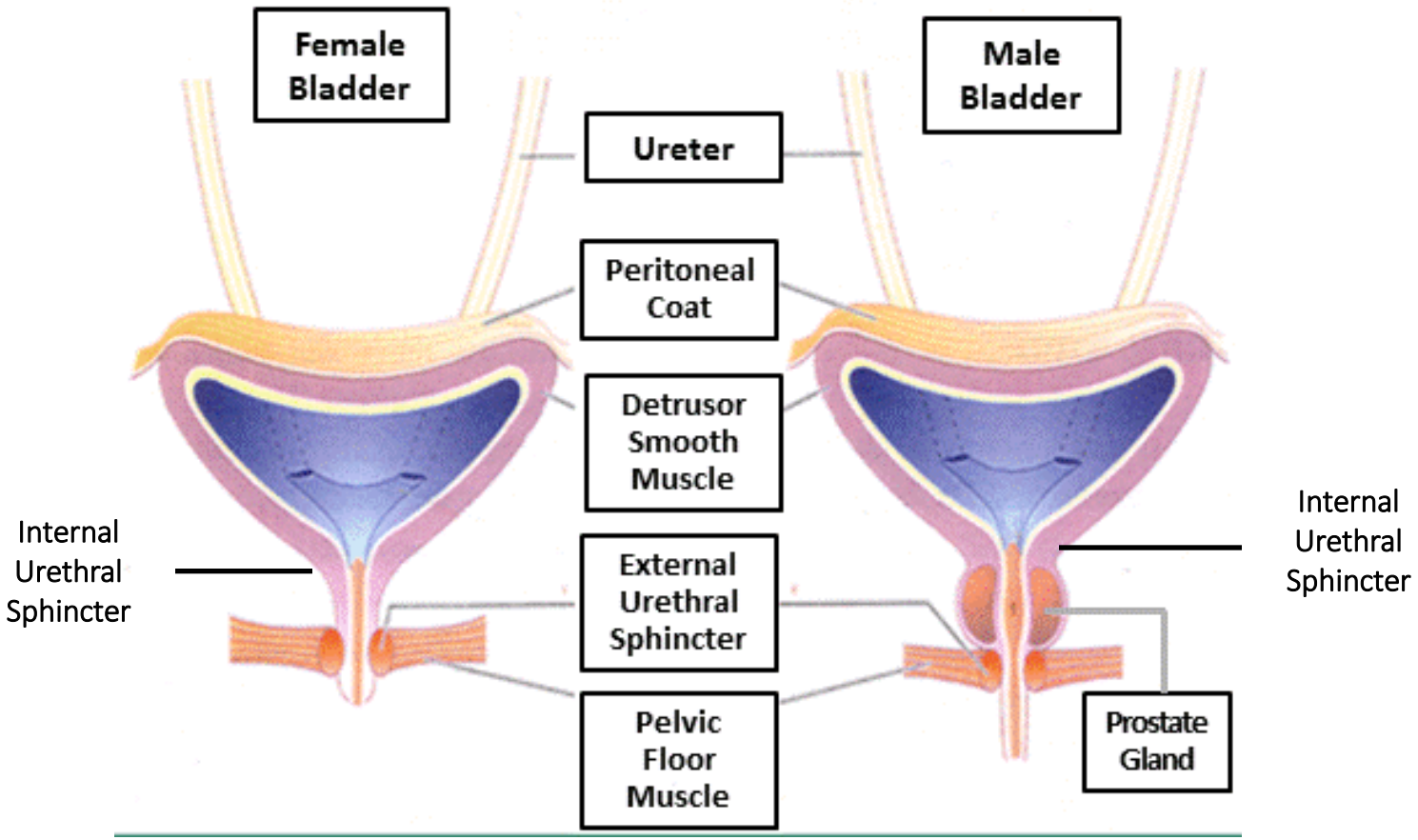
Lower Urinary Tract Anatomy

Urethral sphincters

- Internal urethral sphincter
  - Not a true “sphincter”
  - Smooth muscle
  - Involuntary control
  - Passive continence

- External urethral sphincter
  - Smooth + Striated muscle
  - Involuntary + Voluntary control
  - Passive + Active continence

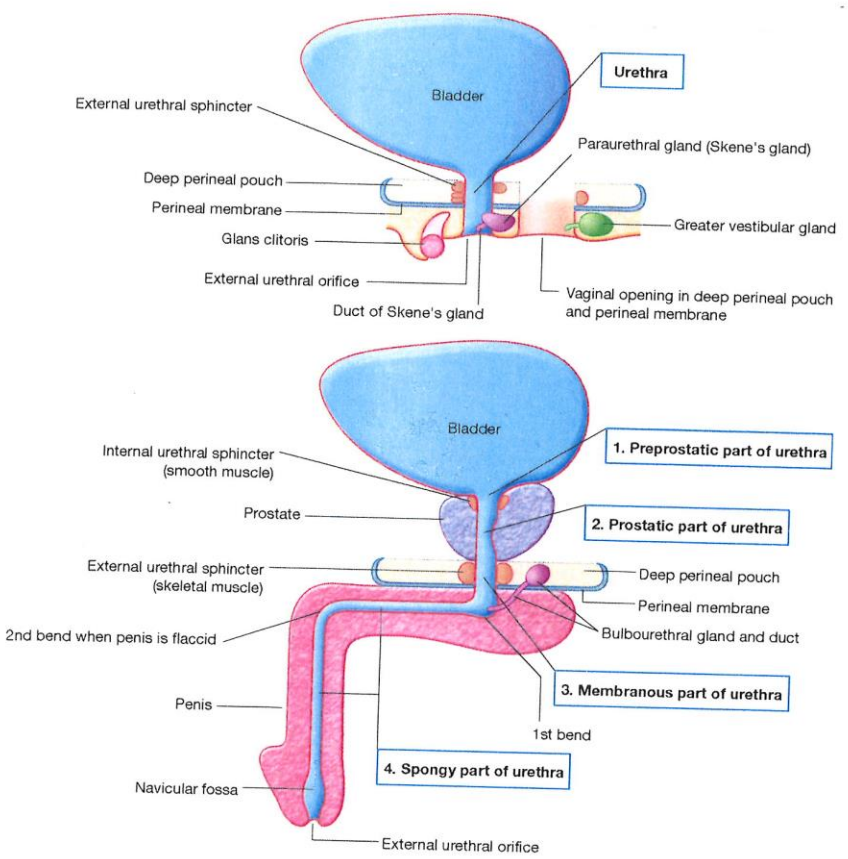
Lower Urinary Tract Anatomy



Lower Urinary Tract Anatomy

- Female urethra
  - 3-5 cm length
  - Short urethra = higher rates of
    1. UTIs
    2. Incontinence

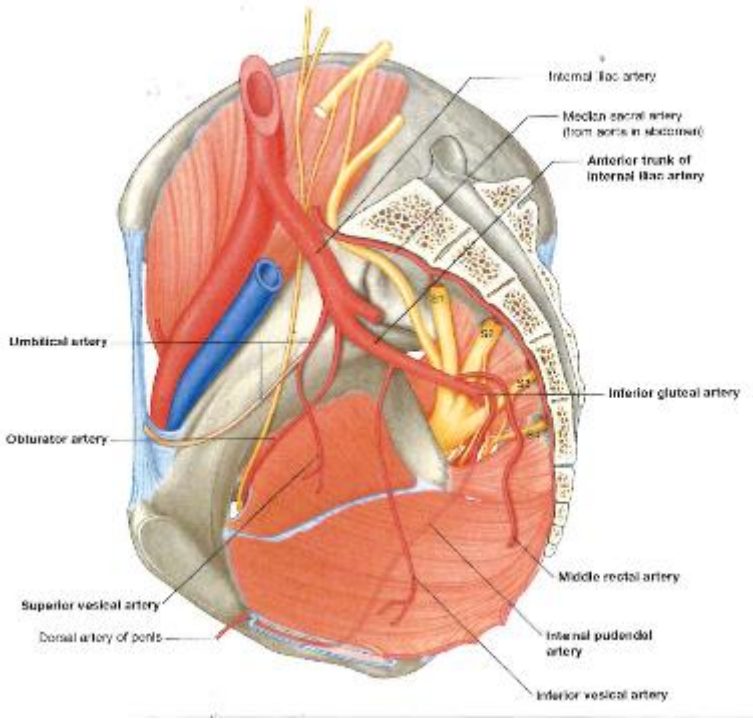
- Male urethra
  - Variable length (20 cm)
  - Long urethra protects against
    1. UTIs
    2. Incontinence
  - Prostatic enlargement



Lower Urinary Tract Anatomy

Bladder Blood Supply

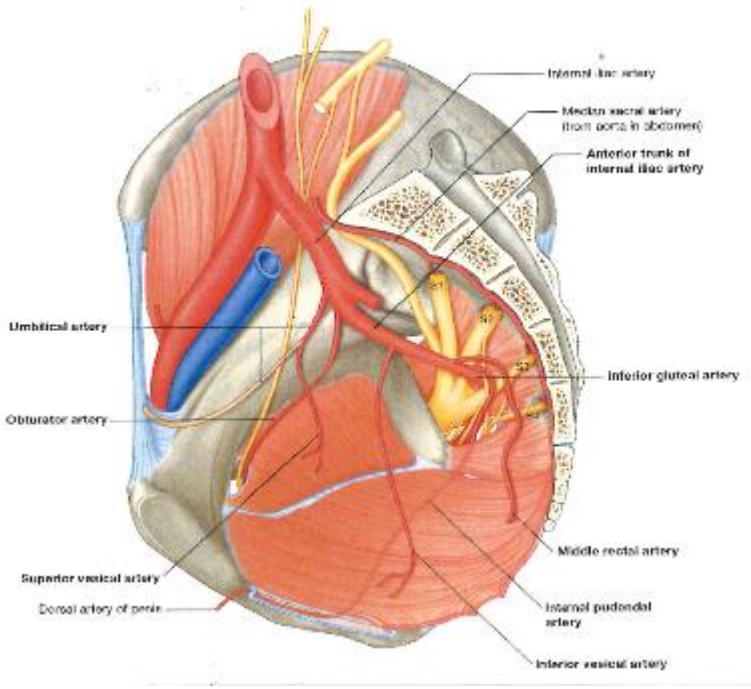
- Anterior trunk of internal iliac A
  - Umbilical A → Superior vesical A
    1. Bladder body
    2. Distal ureter
  - Inferior vesical A
    1. Bladder base
    2. Distal ureter
    3. Prostate
    4. Seminal vesicles
    - Inferior vesical A = Vaginal A



Lower Urinary Tract Anatomy

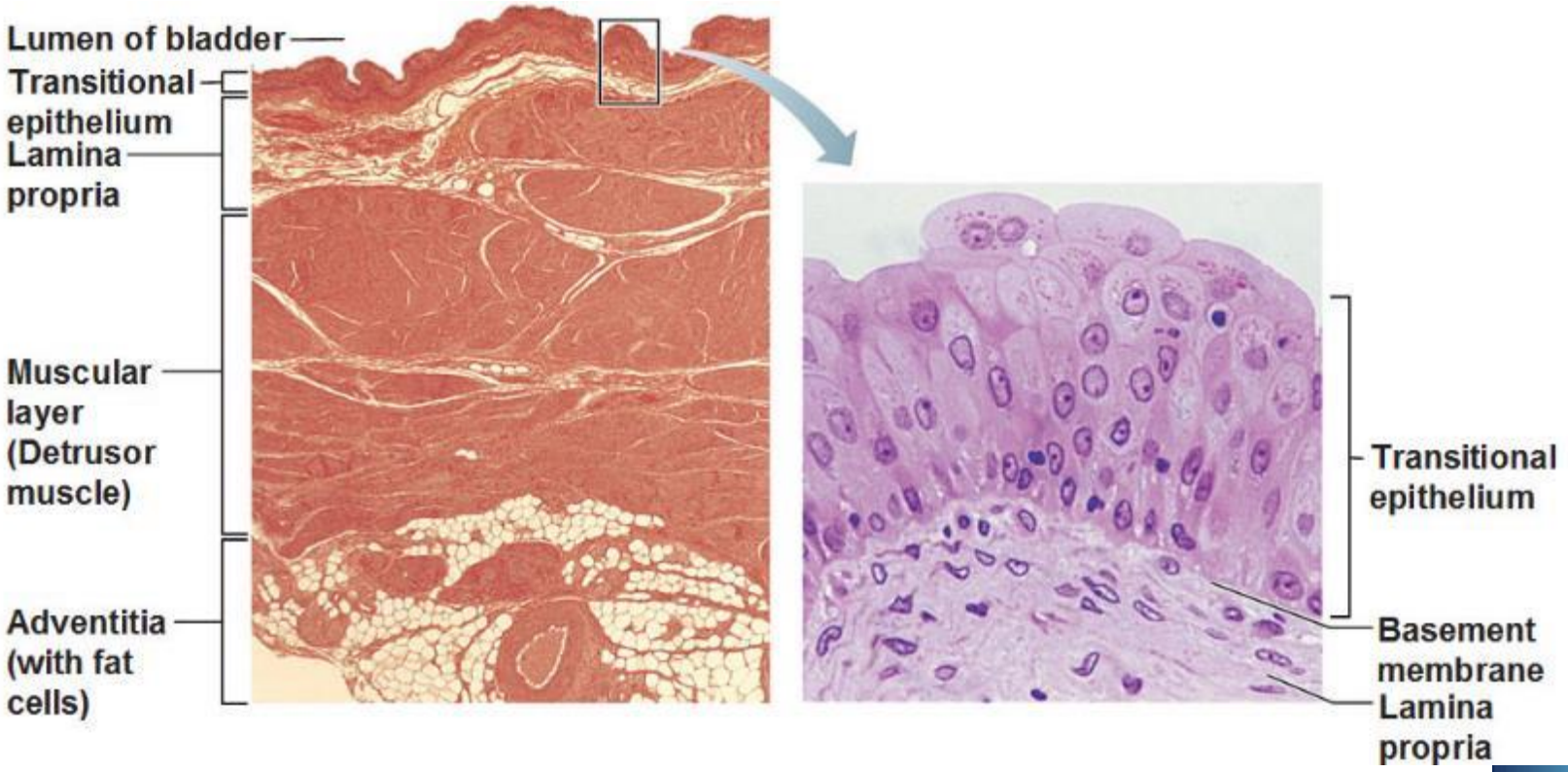
Urethral Blood Supply

- Female
  - Anterior trunk of internal iliac A
    - Internal pudendal A
    - Vaginal A = Inferior vesical A
- Male
  - Anterior trunk of internal iliac A → internal pudendal → Dorsal A of the penis
  - Corpus spongiosum



Lower Urinary Tract Anatomy

Histology



Lower Urinary Tract Anatomy

Histology

- Urothelium = transitional cells
  - 3-7 layers of transitional cells
  - Functions
    - Impermeable barrier to urine
    - Allow for expansion/contraction

- Female urethra
  - Urothelium proximally near bladder neck
  - Squamous epithelium in mid/distal urethra
- Male urethra
  - Urothelium $\rightarrow$ squamous epithelium only at distal end

Normal Lower Urinary Tract Physiology

Function

1. **Storage**
   - Maintain low intravesical pressures
   - Protect upper tracts (kidneys)
   - Maintain continence
   - Provide appropriate warning + sensation

2. **Emptying**
   - Voluntary control
   - Coordination between bladder detrusor muscle + urethral sphincters
   - Empty to completion/near completion

Normal Lower Urinary Tract Physiology

Micturition (Voiding) Reflex

- **First:** voluntary relaxation of striated external urethral sphincter
- **Second:** simultaneous detrusor contraction + urethral smooth muscle relaxation

- Requires integration of
  - Autonomic nervous system (parasympathetic + sympathetic)
  - Somatic nervous system

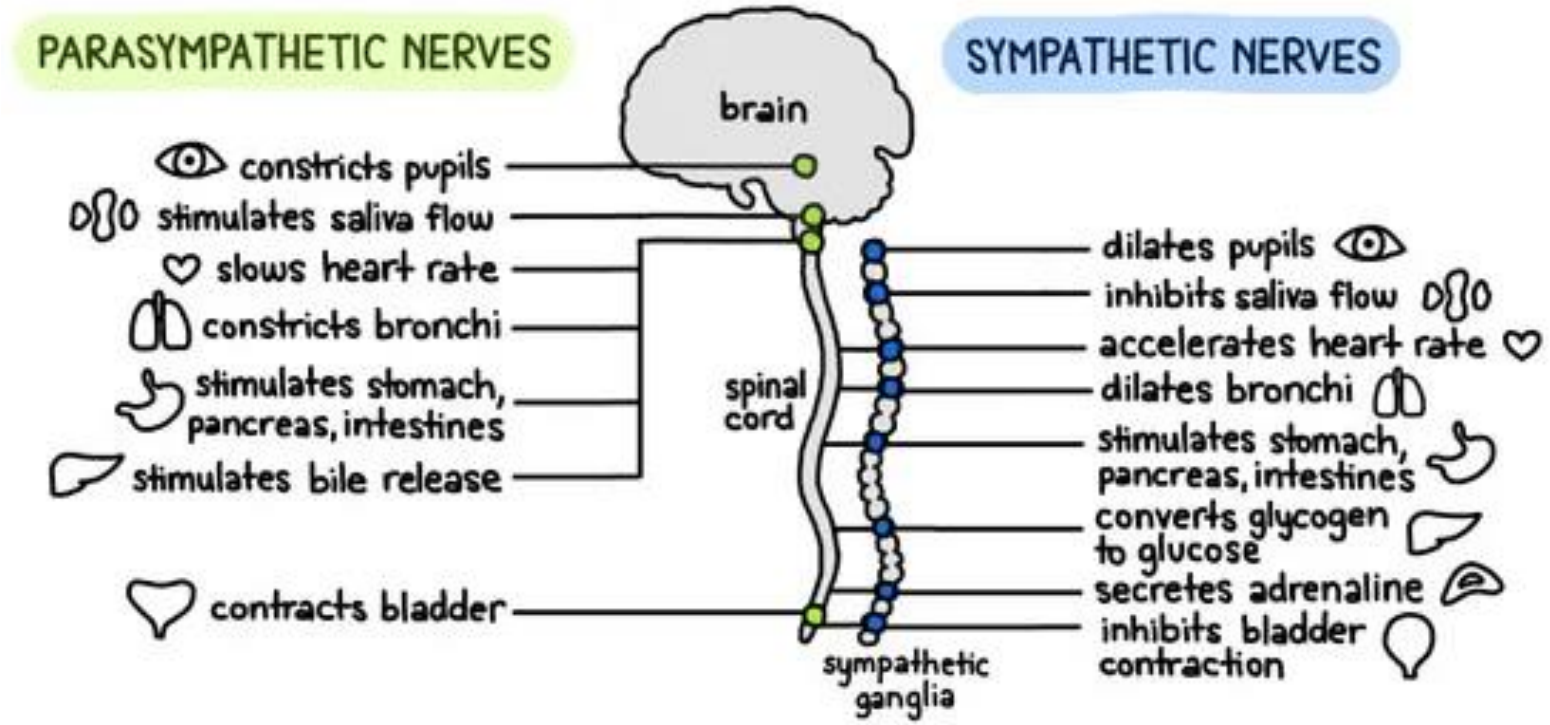
Normal Lower Urinary Tract Physiology

Peripheral Nerves

- **Hypogastic nerve**
  - Sympathetic fibers → sympathetic chain ganglia → pelvic plexus
- **Pelvic nerve**
  - Parasympathetic S2-S4 ventral roots → pelvic plexus
- **Pudendal nerve**
  - S2-S4 ventral roots
  - Somatic
  - Striated urethral + anal sphincters

- *All are mixed motor + sensory nerves*

Normal Lower Urinary Tract Physiology



Normal Lower Urinary Tract Physiology

Sympathetic Innervation

- Mediated by hypogastric nerve
- Releases noradrenaline
  - β-3 adrenergic receptor: relaxes bladder
  - α adrenergic receptor: contracts bladder neck + urethra



Normal Lower Urinary Tract Physiology

Parasympathetic Innervation

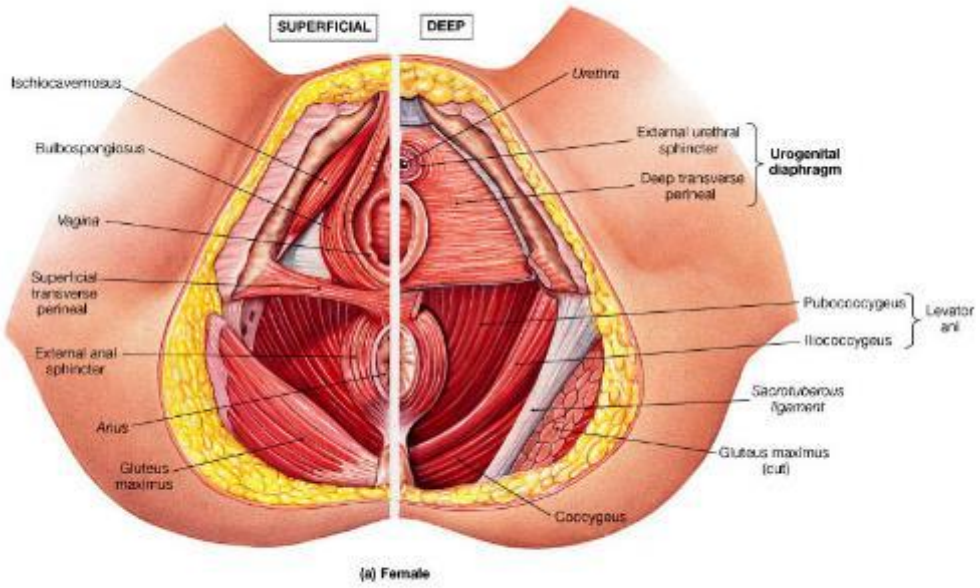
- Mediated by pelvic nerve
- Releases acetylcholine (ACH)
    - ACH receptors: bladder contraction



Normal Lower Urinary Tract Physiology

Somatic Innervation

- Mediated by pudendal nerve
- Pudendal nucleus (Onuf's nucleus) controls striated muscle of external urethral sphincter
- Centrally located motor nucleus controls striated muscle of pelvic floor



Normal Lower Urinary Tract Physiology

Sensory Innervation

- Afferent input conveyed to spinal cord via pelvic + hypogastric nerves
- Aδ fibers
  - Location: detrusor muscle
  - Function: sense bladder fullness (wall tension)
  - Inflammation causes increased discharge at lower threshold



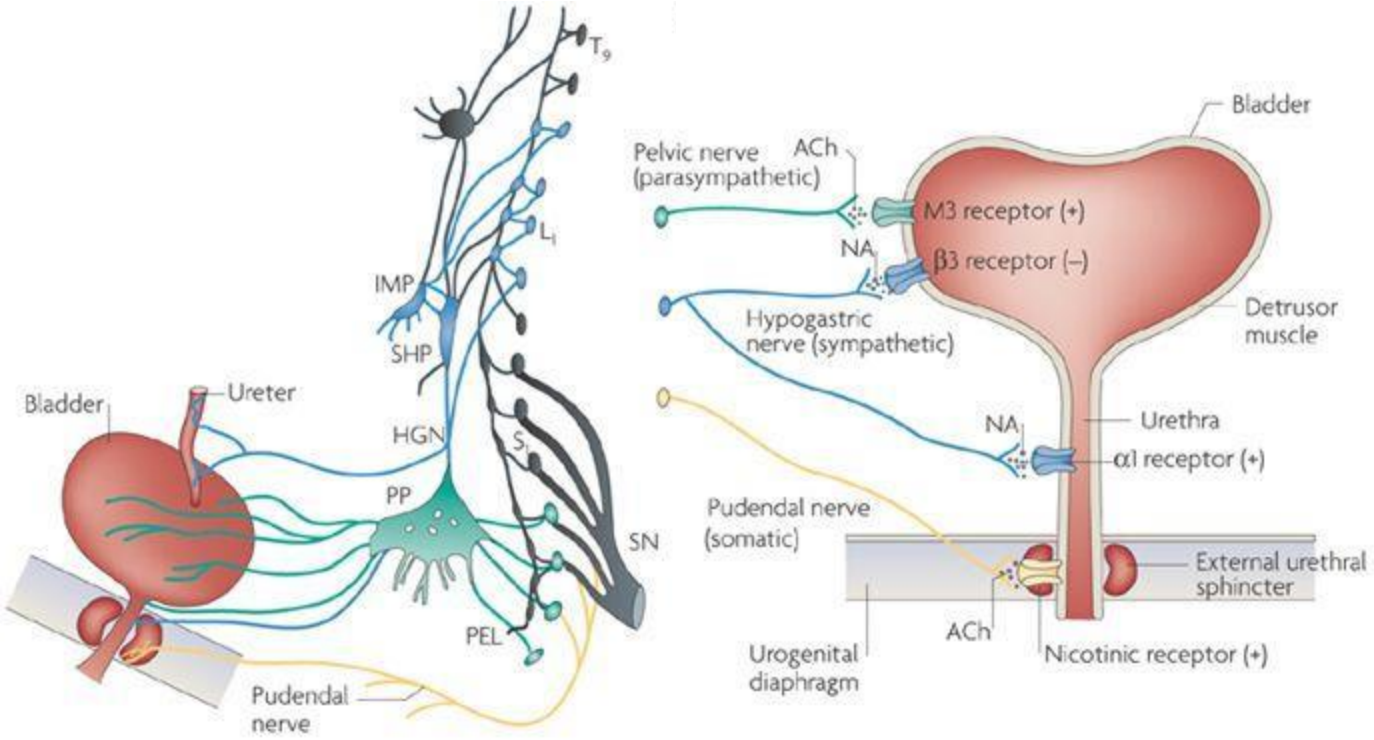
Normal Lower Urinary Tract Physiology

Sensory Innervation

- Unmyelinated C fibers
  - Location: urothelium
  - Function: sense bladder fullness but at higher threshold compared to Aδ fibers → no discharge under normal conditions
    - Senses overdistension
  - Respond to noxious stimuli
    - Chemical irritants, cold, etc



Normal Lower Urinary Tract Physiology



IMP: inferior mesenteric plexus	PP: pelvic plexus
SHP: superior hypogastric plexus	PEL: pelvic nerves
HGN: hypogastric nerve	SN: sciatic nerve
ACh: acetylcholine	NA: noradrenaline

Urinary Incontinence

- Complaint of involuntary loss of urine
- Significant healthcare burden
  - Patient centered factors
    - Negative impact on quality of life
    - Risk factor for infections
    - Dermatologic considerations
  - Economic
    - Direct + indirect costs
    - Estimated $80+ billion cost burden in U.S. in 2020

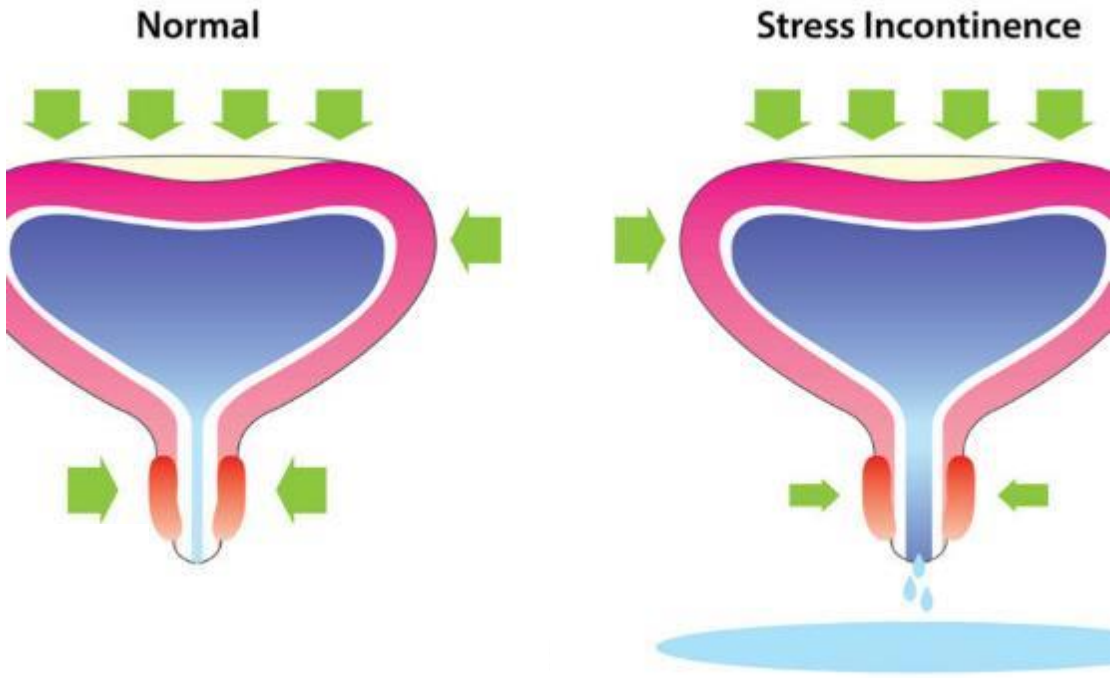
Urinary Incontinence

Epidemiology

- Hard to determine accurate prevalence rates
- Incontinence “normalized” as part of aging process
- Prevalence increases with age
  - Most dramatic rise > 50 years old
- Female >> male
  - > 30% postmenopausal

Stress Urinary Incontinence

- Urinary incontinence with physical activity or Valsalva maneuver (cough/laugh/sneeze)

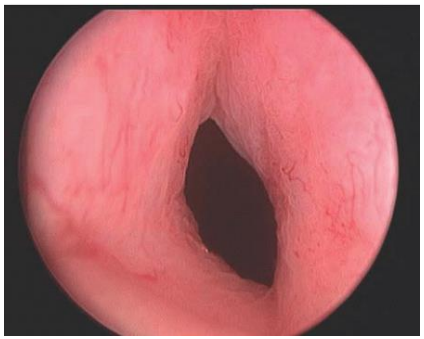


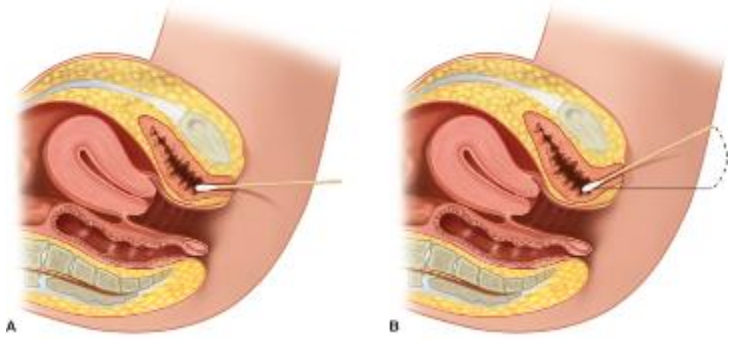
Stress Urinary Incontinence

Pathophysiology

- Loss of urethral resistance to sudden increases in abdominal pressure

- Mechanisms
  - Intrinsic sphincter deficiency: loss of urethral mucosal coaptation
  - Urethral hypermobility: loss of peri-urethral support





Stress Urinary Incontinence

Clinical Presentation

- Urinary incontinence with….
  - Coughing
  - Laughing
  - Sneezing
  - Jumping
  - Running
  - Lifting
  - Bending over



Stress Urinary Incontinence

Risk Factors

- Increased age
- Obesity
- Smoking + chronic cough
- Connective tissue disorder + family history
- Pelvic reconstructive surgery

- Following pregnancy

- Following prostatectomy, prostate reduction surgery

Stress Urinary Incontinence

Diagnosis

- Clinical history
- Physical exam
  - Height + weight → BMI
  - Urinalysis
    - Evaluate for UTI + hematuria
  - Genitourinary exam
    - Cough + Valsalva maneuver
    - Urethral hypermobility (visible only in biologic women)

Stress Urinary Incontinence

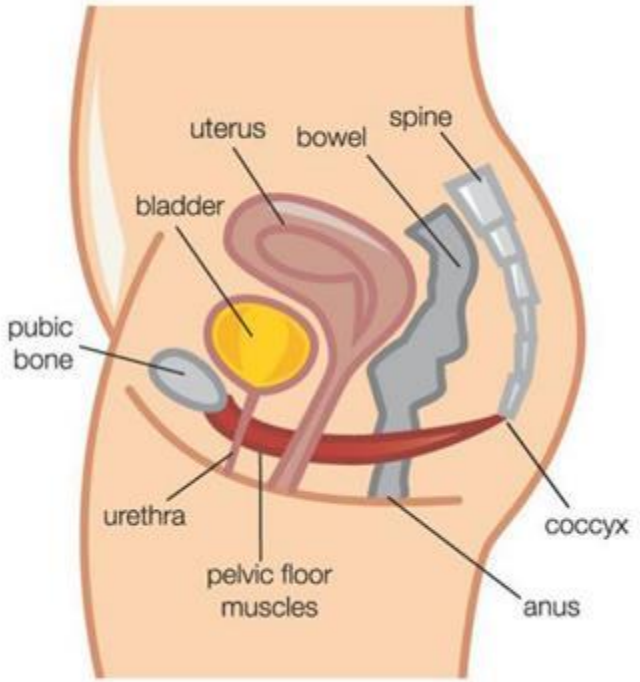
Treatment

- Pelvic Floor Muscle Exercises
  - Kegels
  - Pelvic floor physical therapy
- Peri-urethral Bulking Agents
- Urethral Sling
- Artificial Sphincter for male patients

Stress Urinary Incontinence

Treatment

- Pelvic Floor Muscle Exercises
  - “Kegels”
  - Pelvic floor physical therapy
  - Success rate: ~50%
  - Risks: none
  - 1st line therapy
  - Target patient: motivated, good muscle tone
  - ~40% perform exercises incorrectly → referral to pelvic floor physical therapy

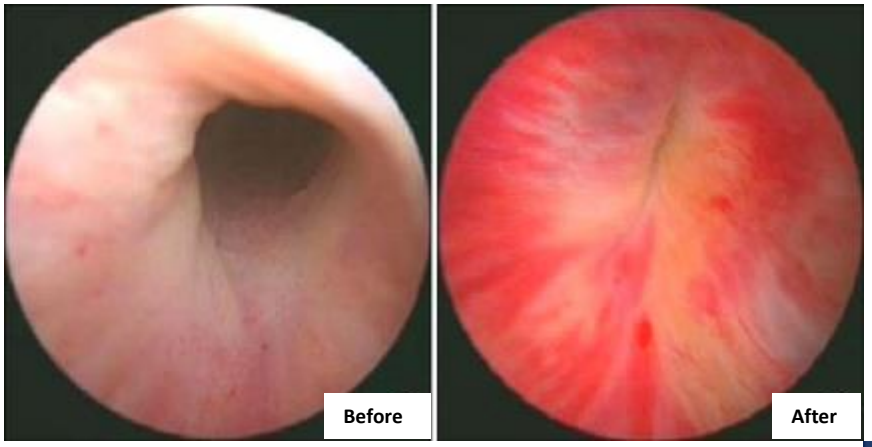


Stress Urinary Incontinence

Treatment

- Peri-urethral Bulking Agents
  - Success rate: 65-80%
  - Risks: transient urinary retention, abscess formation
  - Target patient: mild-moderate stress incontinence

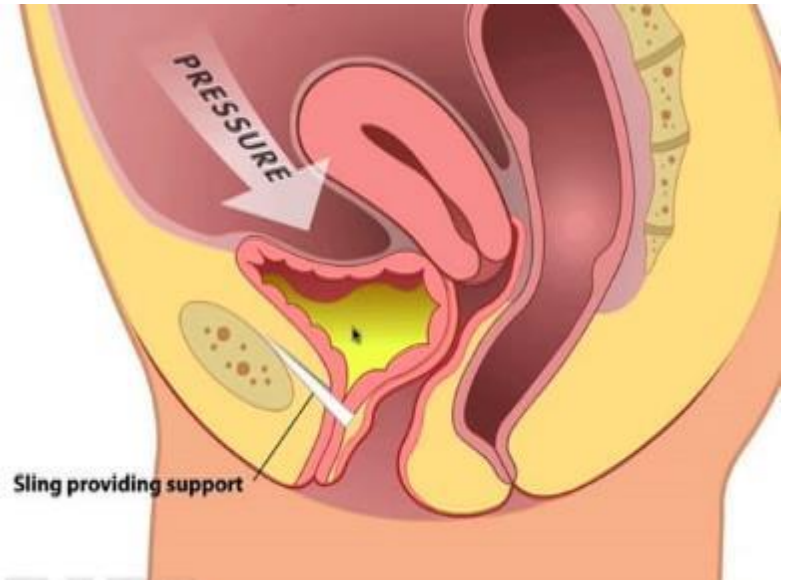




Stress Urinary Incontinence

Treatment

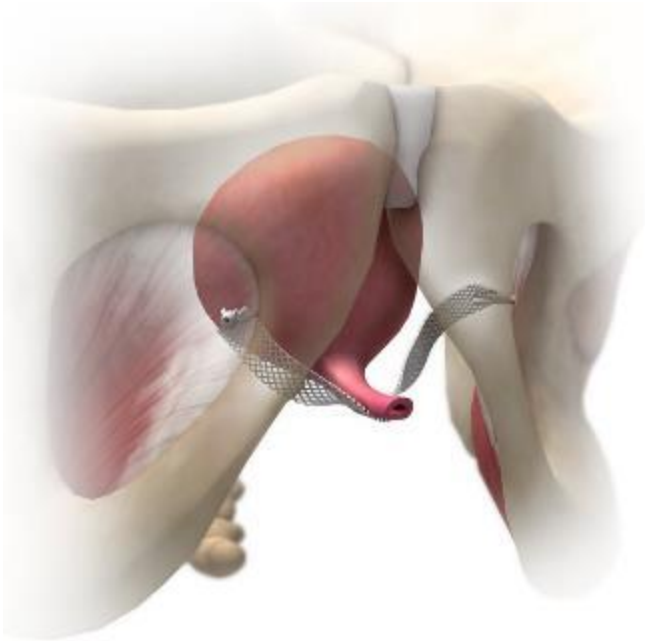
- Urethral sling
  - Fascial: only in female
  - Synthetic (mesh)
  - Hammock of support



Stress Urinary Incontinence

Treatment

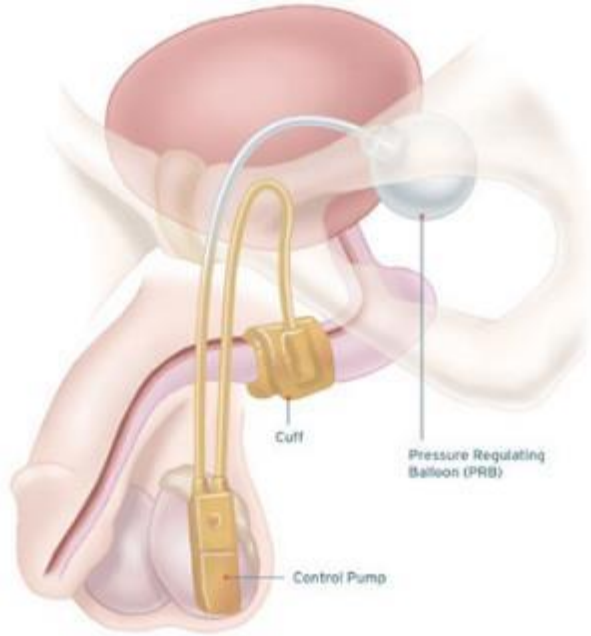
- Synthetic (mesh) sling
  - Success rate:
    - Female 85-90%
    - Male 65-80%
  - Risks:
    - vaginal mesh exposure
    - mesh perforation of urethra/ bladder/bowel
    - urinary retention
    - pelvic pain
  - Target patient: moderate-severe stress incontinence



Stress Urinary Incontinence

Treatment

- Artificial sphincter
  - Males
  - Success rate: 80%
  - Risks:
    - Urethral erosion
    - Infection
    - Mechanical failure
  - Target patient:
    - Moderate-severe stress incontinence
    - Adequate manual dexterity



Overactive Bladder + Urgency Urinary Incontinence

- **Urinary urgency**: sudden compelling desire to urinate which is difficult to defer
  - Usually accompanied by frequency and nocturia ± urgency urinary incontinence
  - Frequency: complaint of voiding more frequently during waking hours than previously deemed “normal”
  - > 8 voids/day

- Urgency urinary incontinence: urinary incontinence associated with sense of urgency

Overactive Bladder + Urgency Urinary Incontinence

Pathophysiology

- Involuntary detrusor contractions
  - “bladder spasms”
- Increased afferent input
  - “sensitive bladder”

- Etiology
  - Idiopathic
  - Neurogenic

Overactive Bladder + Urgency Urinary Incontinence

Clinical Presentation

- Concern regarding voiding frequency
  - Day ± night
- “Can’t make it to the bathroom in time”
- “No warning time”



Overactive Bladder + Urgency Urinary Incontinence

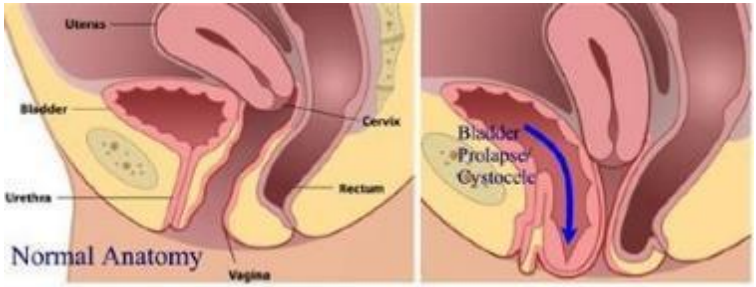
Risk Factors

- Increased age
- Obesity
- Diabetes
- Neurologic factors
  - Multiple Sclerosis, Parkinson's Disease, CVA, etc
- Bladder outlet obstruction

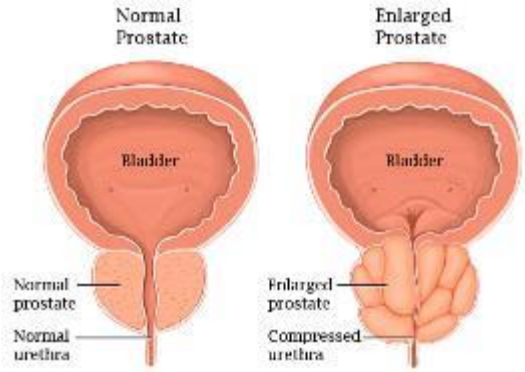
Overactive Bladder + Urgency Urinary Incontinence

Risk Factors

- Bladder outlet obstruction
  - Females: pelvic organ prolapse → urethral kinking



  - Males prostatic enlargement, urethral stricture





Overactive Bladder + Urgency Urinary Incontinence

Diagnosis

- Clinical history
- Physical exam
  - Height + weight → BMI
  - Urinalysis
    - Evaluate for UTI + hematuria
      - Hematuria → evaluate for bladder tumor
  - Genitourinary exam
    - Assess for prolapse
    - Assess for enlarged prostate

Overactive Bladder + Urgency Urinary Incontinence

Treatment

- First-line treatment: behavioral therapy
  - Reduction in bladder irritants
    - e.g. Caffeine, alcohol, artificial sweeteners, acidic beverages, carbonated beverages
  - Fluid management
  - No risks to patients
  - Should be offered to all





Overactive Bladder + Urgency Urinary Incontinence

Treatment

- Second-line treatment: medical therapy
  - Anti-cholinergics
  - β3 agonists

Overactive Bladder + Urgency Urinary Incontinence

Treatment

- Second-line treatment: medical therapy
  - Anti-cholinergics
    - aka “anti-muscarinics”

Trade Name	Generic Name
Vesicare	Solifenacin
Toviaz	Fesoterodine
Sanctura	Trospium
Detrol	Tolterodine
Enablex	Darifenacin
Ditropan	Oxybutynin

Overactive Bladder + Urgency Urinary Incontinence

Treatment

- Second-line treatment: medical therapy
  - Anti-cholinergics
    - Side effects
      - Dry mouth
      - Dry/itchy eyes
      - Constipation
      - Blurred vision
      - Dyspepsia
      - Impaired cognitive function

Overactive Bladder + Urgency Urinary Incontinence

Treatment

- Second-line treatment: medical therapy
  - β3 agonists
    - Mirabegron (Myrbetriq®)
    - Vibegron (Gemtesa®)
    - Side effects
      - Elevated blood pressure
      - Palpitations
      - Nasopharyngitis
      - Headache

Overactive Bladder + Urgency Urinary Incontinence

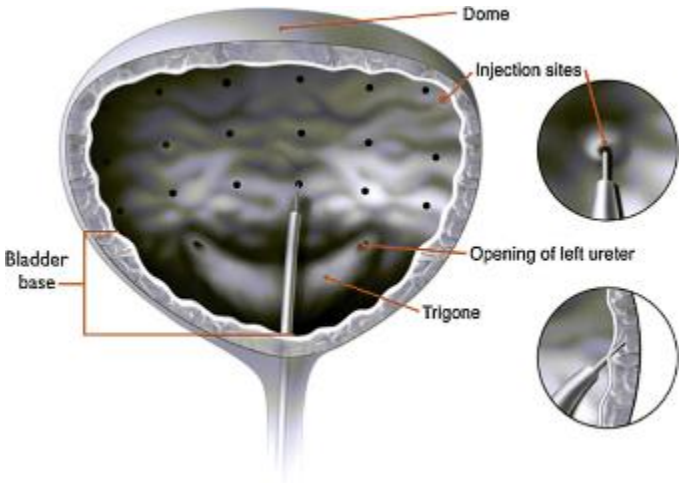
Treatment

- Third-line treatments: various types of bladder neuromodulation
  - Peripheral tibial nerve stimulation
  - Sacral neuromodulation
  - Chemodenervation (onabotulinumtoxinA)

Overactive Bladder + Urgency Urinary Incontinence

OnabotulinumtoxinA (Botox®)

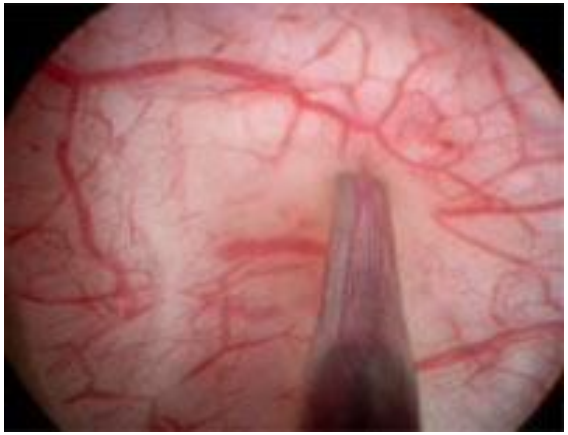
- Most potent neurotoxin known to man
- Blocks presynaptic release of ACH
- Inhibits detrusor contraction (ACH)
- Reversible in 6-12 months



Overactive Bladder + Urgency Urinary Incontinence

OnabotulinumtoxinA (Botox®)

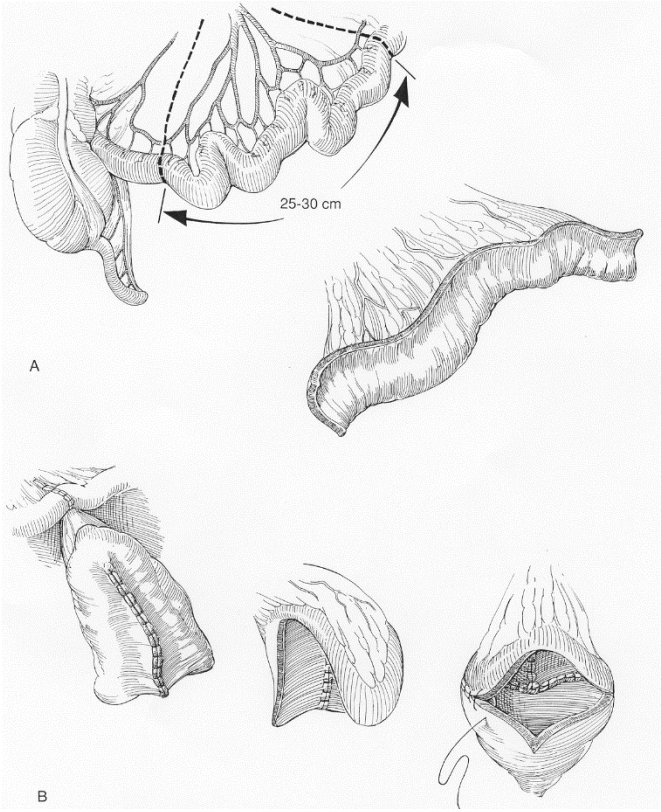
- Success rate: 85%
- Risks:
  - UTI
  - 1-5% Temporary urinary retention
- Target patient:
  - Urgency urinary incontinence
  - Willing to catheterize



Overactive Bladder + Urgency Urinary Incontinence

Treatment

- Fourth-line treatment: augmentation cystoplasty
  - Goals =
    1. Interruption of coordinated detrusor contractions
    2. Increase in bladder capacity
    3. Creation of low pressure bladder
  - Patients must be motivated + willing to catheterize themselves



Neurogenic Bladder

- Dysfunctional lower urinary tract secondary to lesion in brain, spinal cord or peripheral nerves
- Lower urinary tract dysfunction depends on location of lesion
- Acute dysfunction may differ from chronic

Neurogenic Bladder

Lesions **above** the brain stem

- Example: cerebrovascular accident, brain tumor
- Loss of central inhibition
- Clinical presentation:
  - Involuntary detrusor contractions → overactive bladder + urgency incontinence



Neurogenic Bladder

Spinal Cord Lesions

- Example: spinal cord injury, spinal cord tumor
- Spinal shock
  - Initial period of hypocontractile or acontractile detrusor → urinary retention
  - ~ 6 weeks
- Loss of central inhibition
- Clinical presentation:
  - Involuntary detrusor contractions → overactive bladder + urgency incontinence



Neurogenic Bladder

Lower motor neuron injury

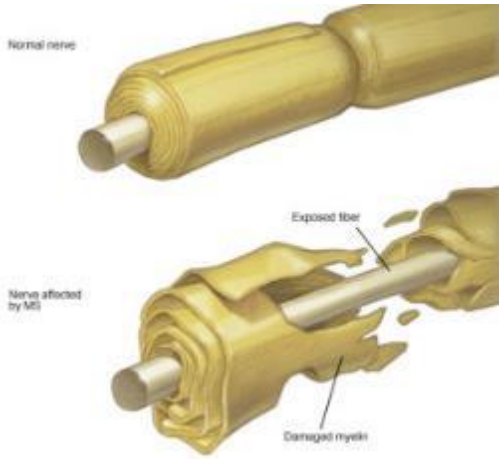
- Example: radical pelvic surgery, cauda equina syndrome
- Clinical presentation:
  - Acontractile detrusor $\rightarrow$ urinary retention



Neurogenic Bladder

Demyelinating lesions

- Example: Multiple sclerosis
- Loss of central inhibition
- Clinical presentation:
  - Involuntary detrusor contractions → overactive bladder + urgency incontinence





Neurogenic Bladder

Diabetes

- Loss of central inhibition
- Impairment of detrusor contractility
- Clinical presentation:
  - Involuntary detrusor contractions → overactive bladder + urgency incontinence
  - Hypocontractile or acontractile detrusor → urinary retention

Bladder Outlet Obstruction

Mechanisms

- Primary bladder neck obstruction
  - Failure of involuntary internal sphincter to relax
  - Treatment: α receptor antagonist

- Dysfunctional voiding
  - Failure of voluntary external sphincter to relax
  - Treatment: biofeedback $\rightarrow$ patients learn to control pelvic floor muscles + coordinate sphincter relaxation with detrusor contraction

Bladder Outlet Obstruction in Females

Mechanisms

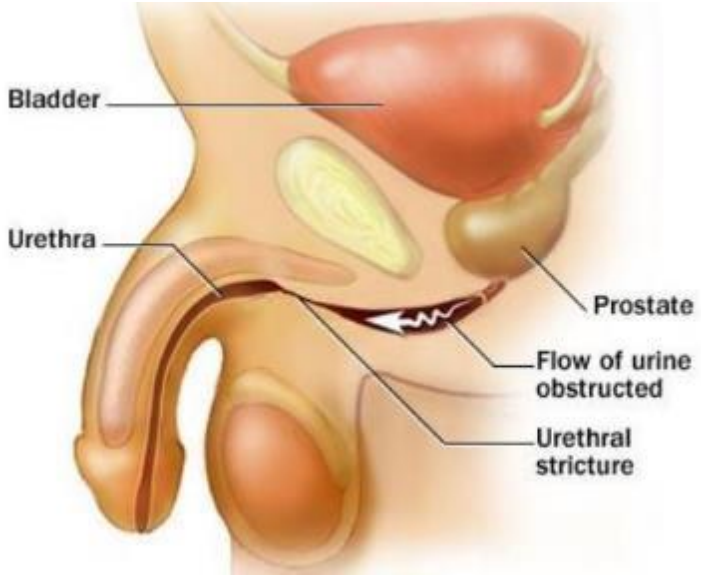
- Pelvic organ prolapse → cystocele
  - Mechanical kinking of bladder neck
  - Treatment: correct cystocele

- Urethral sling too tight
  - Treatment: loosen or remove sling

Bladder Outlet Obstruction in Males

Mechanisms

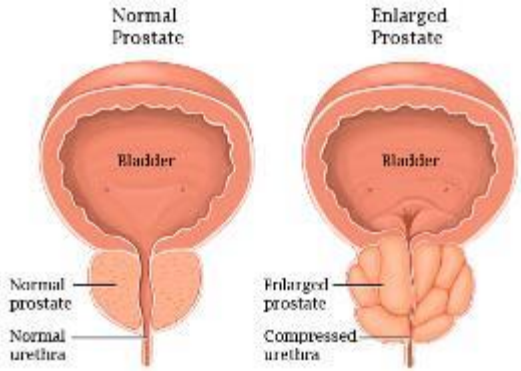
- Urethral stricture
  - Treatment: remove scar tissue, catheter placement



Bladder Outlet Obstruction in Males

Mechanisms

- Prostatic enlargement
  - Benign prostatic enlargement > prostate cancer
  - Treatment:
    - Medical therapy
      - α receptor antagonist
      - Phosphodiesterase-5 inhibitor
      - 5-α reductase inhibitors
    - Prostate reduction procedure (mechanical tissue removal)
    - Prostatectomy for cancer
    - Catheter placement



Overview

- Lower urinary tract anatomy
- Normal lower urinary tract physiology
- Urinary incontinence
  - Stress incontinence
  - Urgency incontinence
- Neurogenic bladder
- Bladder outlet obstruction

Thank You